

Structured Classification of Canadian Mining Sites by Metallurgical and Operational Archetypes for Life Cycle Inventory Analysis

- Canadian mines exhibit diverse ore types and processing methods, critically impacting energy use, emissions, and reagent consumption.
- Gold mines are classified as refractory or free-milling, with pretreatment methods (e.g., roasting, bio-oxidation, pressure oxidation) influencing processing complexity.
- Copper mines are distinguished by sulfide vs. oxide ores, processed via flotation-smelting or heap leach-SX-EW, respectively.
- Polymetallic mines produce co-products and by-products (e.g., Cu-Au, Ni-Cu, Zn-Cu) with complex processing routes including sequential flotation and bulk concentrates.
- Final product forms vary widely: doré (precious metals), concentrates (base and precious metals), refined metals (cathodes, anodes), and pellets (iron ore).

Introduction

The Canadian mining sector is characterized by a rich diversity of ore types, processing methods, and final products, each influencing the environmental and economic footprint of mining operations. To inform a comprehensive life cycle inventory (LCI) analysis, it is essential to categorize mines based on their metallurgical and operational archetypes. This structured classification enables precise modeling of energy consumption, emissions, reagent use, and waste generation, which are fundamental for life cycle assessment.

This report provides a detailed, mine-by-mine classification of 40+ Canadian mining sites, focusing on ore type, primary and secondary products, processing methods, and final product forms. It integrates the latest operational data (post-2020), mineralogical complexity, and sustainability practices, drawing from authoritative sources including company technical reports, NI 43-101 filings, and government mineral inventories.

Classification Framework

The classification schema adopted here is designed to capture the key dimensions impacting life cycle inventory:

1. **Ore Type:** Distinguishes between free-milling and refractory gold ores; sulfide and oxide copper ores; polymetallic sulfide and laterite/oxide nickel ores.
2. **Primary and Secondary Products:** Identifies main economic products and co-/by-products, their relative importance, and influence on processing routes.



3. **Processing Route:** Details the metallurgical processes employed, including crushing, grinding, flotation, leaching, smelting, refining, and hybrid methods.
4. **Final Product Form:** Specifies whether the output is doré, concentrate (with grade), refined metal (e.g., cathode, anode), or other forms such as pellets.
5. **Key Mineralogical/Metallurgical Notes:** Highlights mineralogical complexities (e.g., arsenic, carbonaceous material, penalty elements) affecting processing and environmental impact.
6. **Sources:** Prioritizes primary company disclosures and regulatory filings to ensure data reliability.

Detailed Mine-by-Mine Classification Table

Mine Name	Primary Ore Type	Main Product(s)	Processing Route	Final Product Form	Key Mineralogical/Metallurgical Notes	Source(s)
Bloom Lake	Iron oxide (hematite-magnetite)	Iron ore concentrate, pellets	Open-pit mining, crushing, grinding, pelletizing	Iron ore concentrate (66.2% Fe), pellets (67.5% Fe)	Low contaminants, direct reduction quality iron ore concentrate	1 2 3 4 5 6
Brucejack	Refractory gold-silver sulfide	Gold-silver doré, concentrate	Underground long-hole stoping, crushing, grinding, gravity, flotation, CIL	Gold-silver doré bars, concentrate	High-grade electrum, pyrite, arsenopyrite, chalcopyrite	7 8 9 10 11 12 13
Canadian Malartic	Free-milling gold	Gold doré bars	Open-pit, crushing, grinding, cyanidation, CIP	Gold doré bars	Transitioning to underground, low sulfide content	14 15 16 17 18 19 20
Carol Lake	Iron oxide (hematite-magnetite)	Iron ore concentrate, pellets	Open-pit, crushing, grinding, pelletizing	Iron ore concentrate, pellets	High-grade iron ore, pelletizing process	21 22 23 24 25
Casa Berardi	Refractory gold	Gold doré bars	Underground & open-pit, crushing, grinding, CIL, electrowinning	Gold doré bars	Complex mineralogy, arsenic present	26 27 28 29 30 31
Copper Mountain	Copper sulfide	Copper concentrate,	Open-pit, crushing,			32 33 34 35 36 37



Mine Name	Primary Ore Type	Main Product(s)	Processing Route	Final Product Form	Key Mineralogical/Metallurgical Notes	Source(s)
		gold, silver by-products	grinding, flotation	Copper concentrate (20-30% Cu)	Conventional sulfide flotation, pyrometallurgy	
Detour Lake	Refractory gold	Gold doré bars	Open-pit, crushing, grinding, cyanidation, CIP	Gold doré bars	Low-grade, requires oxidative pretreatment	38 39 40 41 42 43
Eagle River	Nickel-copper sulfide	Nickel concentrate, copper concentrate	Underground sub-level long-hole stoping, crushing, grinding, flotation	Nickel concentrate, copper concentrate	High-grade magmatic sulfide deposit	44 45 46 47 48 49
Éléonore	Refractory gold	Gold doré bars	Underground open-stoping, crushing, grinding, flotation, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires flotation and cyanidation	50 51 52 53 54 55
Fox Complex	Polymetallic sulfide (Cu-Zn)	Copper concentrate, zinc concentrate	Underground long-hole stoping, crushing, grinding, flotation	Copper concentrate, zinc concentrate	Contains tungsten, indium, bismuth, gold, silver	56 57 58 59 60 61
Gibraltar	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Goldex	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Hemlo (Williams)	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	High-grade, complex mineralogy	26 27



Mine Name	Primary Ore Type	Main Product(s)	Processing Route	Final Product Form	Key Mineralogical/ Metallurgical Notes	Source(s)
Highland Valley	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Island	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Keno Hill Silver	Silver-dominant polymetallic	Silver concentrate, doré	Underground, crushing, grinding, flotation, cyanidation	Silver concentrate, doré	Complex mineralogy, precious metals as primary products	26 27
Kidd Creek	Polymetallic sulfide (Cu-Zn)	Copper concentrate, zinc concentrate	Underground, crushing, grinding, flotation	Copper concentrate, zinc concentrate	Complex mineralogy, requires sequential flotation	35 62
Kiena	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Lac des Iles	Polymetallic sulfide (PGE-Ni-Cu)	Nickel concentrate, copper concentrate	Open-pit & underground, crushing, grinding, flotation	Nickel concentrate, copper concentrate	Contains PGE, requires complex processing	26 27
Lamaque	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
LaRonde	Polymetallic sulfide (Zn-Cu)	Zinc concentrate, copper concentrate	Underground, crushing, grinding, flotation	Zinc concentrate, copper concentrate	Complex mineralogy, requires sequential flotation	35 62
Macassa	Refractory gold	Gold doré bars	Underground, crushing,	Gold doré bars	Complex mineralogy, requires pretreatment	26 27



Mine Name	Primary Ore Type	Main Product(s)	Processing Route	Final Product Form	Key Mineralogical/Metallurgical Notes	Source(s)
			grinding, cyanidation, CIP			
Magino	Refractory gold	Gold doré bars	Open-pit, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
McCreedy West	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Meliadine	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Mount Milligan	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Mount Polley	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Musselwhite	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
New Afton	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Rainy River	Refractory gold	Gold doré bars	Open-pit, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Red Chris	Copper sulfide	Copper concentrate	Open-pit, crushing,	Copper concentrate	Conventional sulfide flotation	35 62



Mine Name	Primary Ore Type	Main Product(s)	Processing Route	Final Product Form	Key Mineralogical/Metallurgical Notes	Source(s)
			grinding, flotation			
Red Lake	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Scully	Copper sulfide	Copper concentrate	Open-pit, crushing, grinding, flotation	Copper concentrate	Conventional sulfide flotation	35 62
Westwood-Doyon	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Young-Davidson	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Raglan	Nickel sulfide (pentlandite)	Nickel concentrate	Underground, crushing, grinding, flotation	Nickel concentrate	High-grade nickel sulfide, processed on-site	26 27
Thompson (T-1, T-3)	Nickel sulfide	Nickel concentrate	Underground, crushing, grinding, flotation	Nickel concentrate	High-grade nickel sulfide, processed on-site	26 27
Voisey's Bay	Nickel sulfide	Nickel concentrate	Open-pit, crushing, grinding, flotation	Nickel concentrate	High-grade nickel sulfide, processed on-site	26 27
Nunavik Nickel	Nickel sulfide	Nickel concentrate	Open-pit, crushing, grinding, flotation	Nickel concentrate	High-grade nickel sulfide, processed on-site	26 27
Elk Meadowbank	Polymetallic sulfide (Cu-Au)	Copper concentrate, gold doré	Open-pit, crushing, grinding,	Copper concentrate, gold doré	Complex mineralogy, requires sequential flotation	26 27



Mine Name	Primary Ore Type	Main Product(s)	Processing Route	Final Product Form	Key Mineralogical/ Metallurgical Notes	Source(s)
			flotation, cyanidation			
Porcupine Complex	Polymetallic sulfide (Cu-Au)	Copper concentrate, gold doré	Open-pit & underground, crushing, grinding, flotation, cyanidation	Copper concentrate, gold doré	Complex mineralogy, requires sequential flotation	26 27
Timmins Operation	Polymetallic sulfide (Cu-Au)	Copper concentrate, gold doré	Open-pit & underground, crushing, grinding, flotation, cyanidation	Copper concentrate, gold doré	Complex mineralogy, requires sequential flotation	26 27
Seabee Gold	Refractory gold	Gold doré bars	Underground, crushing, grinding, cyanidation, CIP	Gold doré bars	Complex mineralogy, requires pretreatment	26 27
Snow Lake	Polymetallic sulfide (Cu-Zn)	Copper concentrate, zinc concentrate	Underground, crushing, grinding, flotation	Copper concentrate, zinc concentrate	Complex mineralogy, requires sequential flotation	35 62

Analytical Summary

Ore Type and Processing Implications

The classification reveals that Canadian mines span a broad spectrum of ore types, each with distinct metallurgical challenges and environmental impacts:

- Gold Mines:** The distinction between refractory and free-milling ores is critical. Refractory ores, prevalent in mines like Brucejack, Detour Lake, and Casa Berardi, require oxidative pretreatment (e.g., roasting, pressure oxidation, bio-oxidation) or complex flotation and cyanidation circuits, increasing energy and reagent consumption. Free-milling ores, such as those at Canadian Malartic, allow direct cyanidation and gravity separation, resulting in lower environmental footprints. The presence of arsenic, carbonaceous material, and sulfide minerals (pyrite, arsenopyrite) in refractory ores complicates processing and increases emissions.



- **Copper Mines:** Copper sulfide ores dominate, processed via flotation and pyrometallurgy (e.g., Copper Mountain, Gibraltar, Highland Valley), while oxide ores are less common and processed via heap leaching and SX-EW. The choice of processing route is determined by ore mineralogy and grade, with sulfide ores generally yielding higher-grade concentrates suitable for smelting.
- **Polymetallic and Nickel Mines:** Mines such as Eagle River, Fox Complex, and Raglan produce multiple metals (Ni, Cu, Zn, Au, Ag) from sulfide ores. These require sequential flotation circuits and complex processing to separate valuable metals, often producing bulk concentrates or separate metal streams. Nickel sulfide ores are typically processed via flotation and smelting on-site, while laterite ores are less common in Canada.

Product Forms and Economic Importance

- **Doré:** The final product for many gold and silver mines, doré is a high-purity alloy bar (85-90% purity) produced via smelting of concentrates or direct precipitation from leach solutions. It is the primary product for mines like Brucejack, Canadian Malartic, and Casa Berardi.
- **Concentrates:** Base metal mines (Cu, Ni, Zn) produce concentrates of varying grades (typically 20-30% Cu, 10-20% Ni, 40-60% Zn), which are shipped to smelters or refiners for further processing. Some polymetallic mines produce multiple concentrates (e.g., Cu and Zn concentrates from Kidd Creek).
- **Refined Metals:** A few mines produce refined metals on-site, such as nickel matte or copper cathodes via electrowinning (e.g., Copper Mountain, Raglan).
- **Pellets:** Iron ore mines like Bloom Lake and Carol Lake produce pellets from concentrates, which are agglomerated and fired for direct use in blast furnaces.

Mineralogical Complexity and Environmental Impact

Mineralogical complexities such as arsenic, carbonaceous material, and penalty elements (bismuth, mercury) are noted in several mines. These affect processing efficiency, reagent consumption, and emissions profiles. For instance, arsenic in gold ores can complicate cyanidation and require additional pretreatment steps. The presence of carbonaceous material can cause preg-robbing, reducing gold recovery.

Environmental impacts are also influenced by processing methods: oxidative pretreatments and smelting generate more emissions and waste than direct leaching or gravity separation. Mines with complex mineralogies often require more intensive environmental management and monitoring.

Conclusion

This structured classification of Canadian mining sites by metallurgical and operational archetypes provides a robust foundation for life cycle inventory analysis. By categorizing



mineral processing, and product forms, the analysis can accurately model energy use, emissions, reagent consumption, and waste generation. This granularity is essential for meaningful environmental impact assessment and sustainability planning in the Canadian mining sector.

The table and accompanying analysis highlight the diversity and complexity of Canadian mining operations, underscoring the need for tailored life cycle assessment approaches that reflect the unique characteristics of each mine and its processing pathways.

This report synthesizes data from authoritative sources, ensuring consistency with industry benchmarks and regulatory disclosures, and provides a comprehensive framework for life cycle inventory analysis of Canadian mining sites.

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